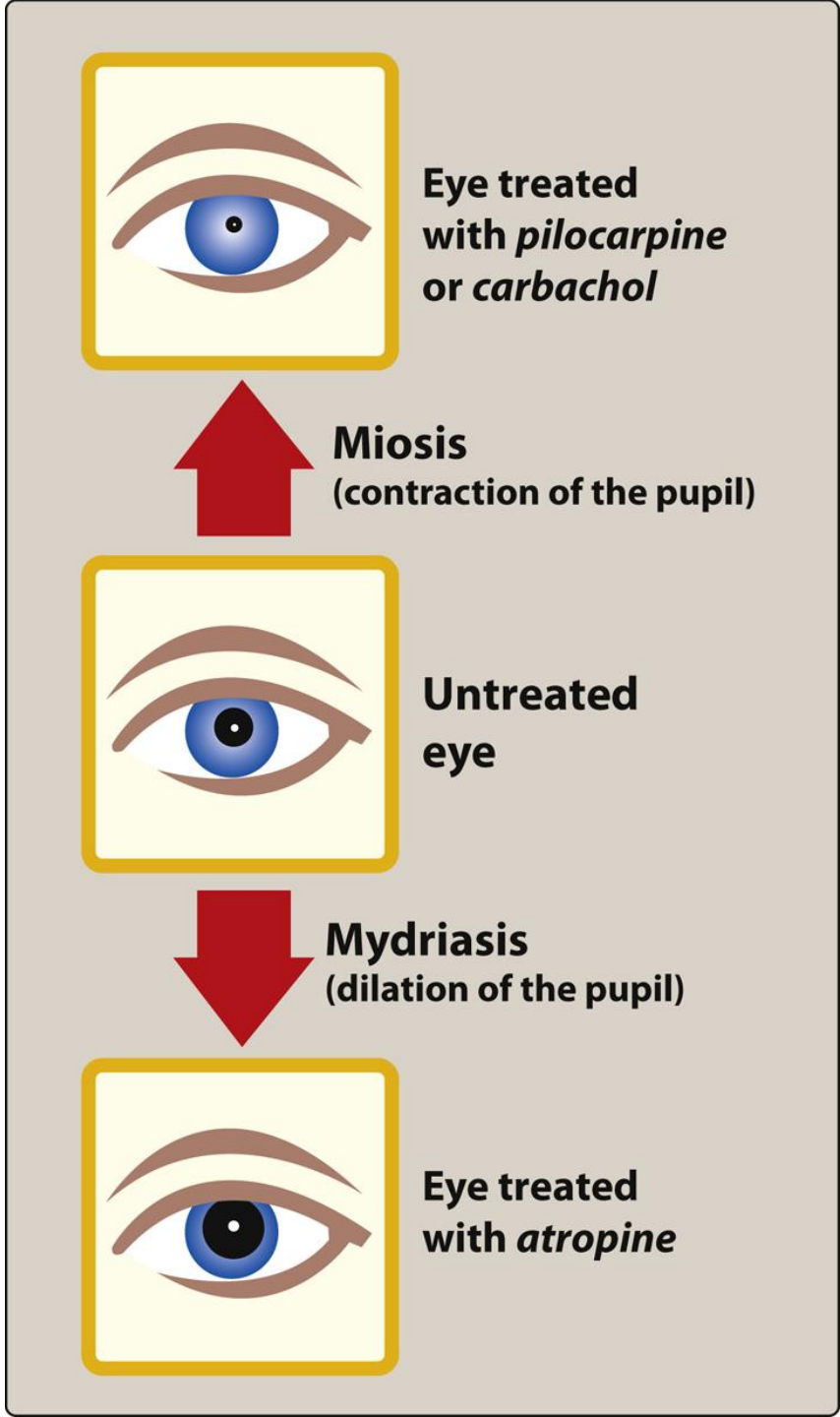


Ophthalmics

Aryan Rahbar, PharmD

Mydriatics & cycloplegics

- Atropine, tropicamide, cyclopentolate
  - Cholinergic antagonists
    - Have mydriatic and cycloplegic activity
  - Primary difference between the agents is duration of mydriasis
    - Atropine: 7-14 days
    - Cyclopentolate: 24 hours
    - Tropicamide: 6 hours
- Phenylephrine
  - α_1 agonist → mydriasis (no cycloplegia)



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Antiglaucoma agents



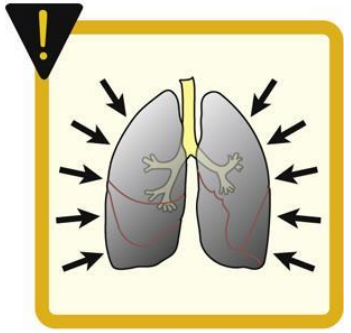
β-blockers (tim**olol**, betax**olol**, carte**olol**)

- Mechanism of action
  - Blocks β_2 activated aqueous humor production from the ciliary epithelium
- Adverse effects
- Indicated for angle closure and open angle glaucoma

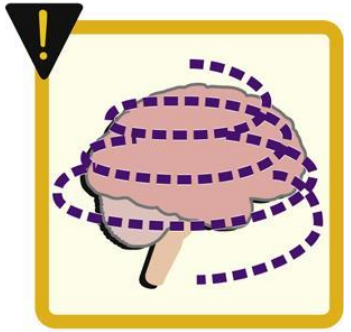
Fatigue



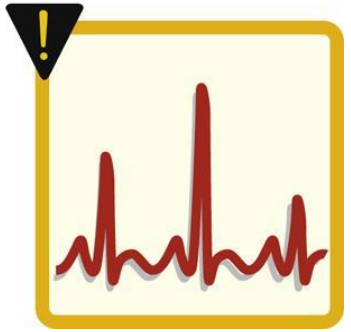
Broncho-constriction



Dizziness



Arrhythmias (upon abrupt withdrawal)



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α-agonists

- α_2 agonist (apracl**onidine**, brim**onidine**)
  - Mechanism of action
    - Primary: ↓ aqueous humor synthesis via α_2 receptor activation
    - Secondary: ↑outflow of aqueous humor through the uveoscleral pathway
  - Adverse effects
    - Blurry vision
    - Ocular hyperemia
    - Foreign body sensation
    - Itchy eyes
  - Indicated for open angle glaucoma

- α and β agonist (epinephrine)
  - Mechanism of action
    - Early effect: ↑ β_2 aqueous humor production
      - DO NOT GIVE IN ANGLE-CLOSURE GLAUCOMA
    - Later effect: ↓ aqueous humor synthesis via vasoconstriction and α_2 receptor activation
    - Later effect: ↑outflow of aqueous humor through the uveoscleral pathway
  - Adverse effects
    - **Mydriasis**
    - Blurry vision
    - Ocular hyperemia
    - Foreign body sensation
    - Itchy eyes
  - Not routinely used anymore for glaucoma management

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Diuretics (acetazolamide, mannitol)

- Carbonic anhydrase inhibitors (aceta**zolamide**, dor**zolamide**, brin**zolamide**)
  - Mechanism of action
    - Inhibits carbonic anhydrase on the ciliary epithelium of the eye → decreased production of aqueous humor
  - Adverse effects
    - Metabolic acidosis
    - Potassium depletion
    - Paresthesia
    - Renal stones
  - Acetazolamide given orally or intravenously → increased risk for systemic effects
  - Indicated for angle closure glaucoma or open angle glaucoma
- Mannitol
  - Mechanism of action
    - ↑ serum osmolarity → draws fluid out of the anterior chamber of the eye
  - Adverse effects
    - Dehydration → prerenal acute kidney injury
  - Indicated for angle closure glaucoma

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Prostaglandin analogs (bimat**oprost**, latan**oprost**)

- Mechanism of action
  - $PGF_{2\alpha}$ analog → increase outflow of aqueous humor via uveoscleral pathway
- Adverse effects
  - Darkening of iris
  - Eyelash growth (Latisse)
- Indicated for open angle glaucoma



Latisse.com
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Cholinomimetics

- Direct (pilocarpine, carbachol)
  - Mechanism of action
    - M_3 receptor activation contracts ciliary muscle → increased outflow of aqueous humor through trabecular meshwork
  - Adverse effects
    - Cholinergic toxicity
    - Miosis → blurry vision in low light
  - Angle closure glaucoma or open angle glaucoma
- Indirect (echothiophate)
  - Mechanism of action
    - Inhibits acetylcholinesterase → increased acetycholine → M_3 receptor activation contracts ciliary muscle → increased outflow of aqueous humor through trabecular meshwork
  - Adverse effects
    - Cholinergic toxicity
    - Miosis → blurry vision in low light
  - Open angle glaucoma

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Rho kinase inhibitors (netar**sudil**, ripa**sudil**)

- Mechanism of action
  - ↓ cell stiffness of trabecular meshwork → increased drainage of aqueous humor
  - Can ↓ fibrosis of trabecular meshwork
- Well tolerated
- Used in open angle glaucoma

RHOPRESSA® (netarsudil ophthalmic solution) 0.02% reduces IOP by increasing the outflow of aqueous humor through the trabecular meshwork[4]

The exact mechanism is unknown.

rhopressa.myalcon.com

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CLASS OF DRUG	DRUG NAMES	MECHANISM OF ACTION	SIDE EFFECTS
β-Adrenergic antagonists (topical)	*Betaxolol*, *carteolol*, *levobunolol*, *timolol*	Decrease of aqueous humor production	Ocular irritation; contraindicated in patients with asthma, obstructive airway disease, bradycardia, and congestive heart failure.
α-Adrenergic agonists (topical)	*Apraclonidine, brimonidine*	Decrease of aqueous humor production and increase of aqueous outflow	Red eye and ocular irritation, allergic reactions, malaise, and headache.
Cholinergic agonists (topical)	*Pilocarpine*, *carbachol*	Increase of aqueous outflow	Eye or brow pain, increased myopia, and decreased vision.
Prostaglandin-like analogues (topical)	*Latanoprost, travoprost, bimatoprost*	Increase of aqueous humor outflow	Red eye and ocular irritation, increased iris pigmentation, and excessive hair growth of eye lashes.
Carbonic anhydrase inhibitors (topical and systemic)	*Dorzolamide* and *brinzolamide* (topical), *acetazolamide,* and *methazolamide* (oral)	Decrease of aqueous humor production	Transient myopia, nausea, diarrhea, loss of appetite and taste, and renal stones (oral drugs).

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Question

A 53-year-old woman suffering from open-angle glaucoma was prescribed pilocarpine to help control her intraocular pressure. The therapeutic effect of pilocarpine is due to:

A. Inhibition of carbonic anhydrase

B. Inhibition of acetylcholinesterase

C. Activation of muscarinic M_3 receptors

D. Blockade of β-adrenergic receptors

Question

If an ophthalmologist wants to dilate the pupils for an eye examination, which of the following class of drugs would be most useful?

A. Muscarinic receptor agonist

B. Muscarinic receptor antagonist

C. Beta-blocker

D. Osmotic diuretic

Questions?

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